

HW 1 – Goodness of Fit Test

Use the following situation for problems 1 – 3:

The top four General Mills cereal brands are Honey Nut Cheerios, Lucky Charms, Trix, and Cookie Crisp. When given those four options, the researchers want to see if any of those four brands are more popular than the others. A random sample of 100 people who purchased one of the four cereals was taken. Here are the results.

Brand	Cheerios	Lucky Charms	Trix	Cookie Crisp
Frequency	33	26	24	17

_____ 1) Which of the following would be an appropriate null hypothesis for the researchers to test?

- (A) $\mu_1 = \mu_2 = \mu_3 = \mu_4$
- (B) The observed counts are all equal to 25.
- (C) Flavor preferences are evenly distributed across the four brands.
- (D) At least one of the four brand preferences is different from the other three.
- (E) The observed counts are equal to the expected counts.

_____ 2) What is the appropriate number of degrees of freedom for a goodness-of-fit test?

- (A) 25
- (B) 99
- (C) 24
- (D) 100
- (E) 3

_____ 3) Is there sufficient evidence at the 5% significance level that the customers have a brand preference?

- (A) No, since with the chi-square statistic equal to 5.2, there is not sufficient evidence at $\alpha = 0.05$.
- (B) Yes, since with the chi-square statistic equal to 5.2, there is sufficient evidence at $\alpha = 0.05$.
- (C) No, since with the chi-square statistic equal to 5.78, there is not sufficient evidence at $\alpha = 0.05$.
- (D) Yes, since with the chi-square statistic equal to 5.78, there is sufficient evidence at $\alpha = 0.05$.
- (E) No, since with the chi-square statistic equal to 5.78, there is not sufficient evidence at $\alpha = 0.05$.

_____ 4) The P-value for a chi-squared goodness-of-fit test is

- (A) the area to the left of the calculated value of χ^2 under the chi-squared curve.
- (B) the area to the right of the calculated value of χ^2 under the chi-squared curve.
- (C) twice the area to the left of the calculated value of χ^2 under the chi-squared curve.
- (D) twice the area to the right of the calculated value of χ^2 under the chi-squared curve.
- (E) the area between the positive and negative value of χ^2 under the chi-squared curve.

5) Are more babies born on a specific day of the week than others? To determine if the distribution of births across the week happened in different proportions than expected, a researcher took a random sample of 84 births in the year from a local hospital and recorded what day they were born on. The data is given in the following table:

Day of the week	Sun	Mon	Tue	Wed	Thu	Fri	Sat
# of births	6	7	9	14	19	17	12

Based on these data, is it reasonable to conclude that the proportion of births is not the same for all days of the week? Use $\alpha = 0.05$.